

HARSHAL KOLHE

 [GitHub](#) |  [LinkedIn](#) |  harshalkolhe04@gmail.com

EDUCATION

Visvesvaraya National Institute of Technology, India

Bachelor of Technology in Chemical Engineering, Minor in Internet of Things

Aug' 23 – May' 27

CGPA: 8.66/10.0

EXPERIENCE

Indian Institute of Technology, Roorkee

May' 26 – Jul' 26

Summer Research Intern | [\[Certificate\]](#)

- Designed a custom secure drone authentication protocol leveraging V2X device communications to enable secure inter-drone communication and authentication.
- Implemented a low-latency communication protocol in Embedded C on the Unex SOM-352 chipset, implementing IEEE 802.11 standards to enable secure authentication and data sharing with 10 ms communication latency.
- Validated the end-to-end communication and authentication system onboard two self-assembled UAVs, demonstrating a successful real-world proof of concept for secure low-latency inter-UAV communication.

Indian Institute of Technology, Roorkee

May' 25 – Jul' 25

Summer Research Intern | [\[Certificate\]](#)

- Developed a reinforcement learning-based control system for a quadcopter carrying a suspended payload, with active suppression of 85% payload-induced oscillations during flight.
- Created a high-frequency data collection and processing pipeline that synchronized 20+ flight and 8+ payload unit data streams for reinforcement learning policy validation.
- Built a processing pipeline for real-time policy deployment on quadcopter hardware, generating a 3K+ sample dataset across 50+ flight experiments.

PROJECTS

CareFlow — Clinic Management System | [\[GitHub\]](#)

Mar' 26 – Apr' 26

- Engineered a full-stack clinic management platform with 38 REST API endpoints, structuring backend logic across 5 route and 5 service modules for patient, doctor, appointment, billing, and administrative workflows.
- Designed an 11-table Oracle SQL database with 3 PL/SQL automation triggers for doctor allocation, automated billing, and payment-rule enforcement.
- Implemented 4 analytical database views and role-based backend workflows to generate insights into revenue, appointments, doctor workload, and medication trends.

Real-Time Guest Management & QR Event Check-in System | [\[GitHub\]](#)

Jan' 26 – Feb' 26

- Architected a real-time guest management system with 7 REST API endpoints using Node.js, Express, and MongoDB to manage guest registration, QR generation, check-in, check-out, email delivery, and deletion workflows.
- Integrated UUID-based tokens, QR-code generation, and SMTP email services to automate the generation and distribution of unique digital credentials for event guests.
- Enabled real-time check-in/check-out synchronization using Socket.IO, powering live guest status and attendance updates across 6 application pages and reusable dashboard components.

PUBLICATIONS

- A. Sharma, M. Ravikiran, S. Chakrabarty, R. Saluja, M. Limaye, **H. Kolhe**, and S. Samiron, "Prompted to Fly: Translating Free-Form Instructions into Schema-Constrained Mission Generation for UAVs using LLMs," *2025 IEEE International Conference on Big Data (BigData)*, Macau, China, 2025, pp. 7278-7286, doi: [10.1109/BigData66926.2025.11401432](https://doi.org/10.1109/BigData66926.2025.11401432).

SKILLS

Languages: C, C++, Python, SQL

Softwares and Tools: FastAPI, Node.js, Express.js, MongoDB, PostgreSQL, Oracle SQL, Docker, Git, GitHub Actions, AWS

Libraries: REST APIs, Socket.IO, JWT, WebSockets, NumPy, Pandas

EXTRACURRICULAR ACTIVITIES

- Presented [ATLAS: A Cooperative Aerial System for Large-Scale Search and Delivery](#) at the ASME's IMECE, Chennai.
- Secured the [Best Paper Presentation Award](#) at the ICAMMS Conference for presenting research work on autonomous systems.
- Secured [First Runner-up](#) prize at the Drone Helix Competition, SVNIT Surat.
- Vice Chairman of IvLabs, Robotics and Artificial Intelligence Lab at VNIT, Nagpur.